VEHICLE BODY OR CLUTCH PEDAL VIBRATES/SHIFT OPERATION DIFFICULTY/NOISE HEARD [F65M-R]

id050335195500

Troubleshooting item	Vehicle body or clutch pedal vibrates/shift operation difficulty/noise heard
Description	• Vehicle body vibrates during idling or while driving • Clutch pedal vibrates • Shift operation is difficult because clutch does not fully release • Teeth grinding noise is heard • Booming noise is heard
Possible cause	**Note** • The vehicle body may vibrate due to poor clutch operation or if the engine speed decreases below the idle speed by clutch operation when accelerating from a standstill, but this does not indicate a vehicle malfunction. • Resonance with engine vibration (mainly with exhaust system parts) • Resonance of rotating parts on vehicle (drive shaft, tire, etc.) • Clutch fluid malfunction • Clutch pedal malfunction • Manual transaxle shift mechanism malfunction • Manual transaxle oil malfunction • Engine mount malfunction • Clutch master cylinder malfunction • Clutch release cylinder malfunction • Clutch unit malfunction • Manual transaxle internal malfunction
System wiring diagram	• Not applicable
Connector diagram	• Not applicable

Diagnostic Procedure

Step	Inspection	Results	Action
1	**VERIFY MALFUNCTION SYMPTOM** • Is the malfunction symptom at the same level compared to a vehicle of the same model?	Yes	Go to repair completion verification.
		No	Go to the next step.
2	**DETERMINE IF MALFUNCTION CAUSE IS CLUTCH OR MANUAL TRANSAXLE** • Install tires of another brand. **Caution** **• When tires of another brand are installed, the same malfunction can occur due to conditions different from customer explanation. Be careful not to confuse it with customer complaint.** **• When installing tires of a different brand, install 4 tires with the same tire wear.** • Perform simulation driving at the same engine speed, gear position, and vehicle speed that customer vehicle had malfunction. • Does the same malfunction occur?	Yes	Reinstall customer's vehicle tire. Go to next step.
		No	The clutch and manual transaxle are normal. (Resonance with engine vibration or resonance with rotating parts of vehicle) If required, determine the cause.
3	**INSPECT IF MALFUNCTION CAUSE IS INSUFFICIENT CLUTCH FLUID LEVEL** • Inspect the clutch fluid. (See CLUTCH FLUID INSPECTION [F65M-R].) • Is the clutch fluid normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
4	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH FLUID LEAKAGE** • Inspect for clutch fluid leakage. (See CLUTCH FLUID LEAKAGE INSPECTION [F65M-R].) • Is there clutch fluid leakage?	Yes	Repair or replace the malfunctioning location and perform the repair completion verification.
		No	Go to the next step.

Step	Inspection	Results	Action
5	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH PEDAL OR TRAPPED AIR IN CLUTCH FLUID** • Inspect the clutch pedal. (See CLUTCH PEDAL INSPECTION [F65M-R].) • Are all the items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
6	**INSPECT IF MALFUNCTION CAUSE IS MANUAL TRANSAXLE SHIFT MECHANISM** • Inspect the following of the manual transaxle shift mechanism. (See MANUAL TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION [F65M-R].) — Installation condition of part — Damage, deterioration, and wear — Shift lever movement • Are all the items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
7	**INSPECT IF MALFUNCTION CAUSE IS INSUFFICIENT MANUAL TRANSAXLE OIL LEVEL** • Inspect the manual transaxle oil. (See MANUAL TRANSAXLE OIL INSPECTION [F65M-R].) • Is the manual transaxle oil normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
8	**INSPECT IF MALFUNCTION CAUSE IS ENGINE MOUNT** • Inspect the following of the engine mount. (See ENGINE MOUNT DISASSEMBLY/ ASSEMBLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — Installation condition of part — Damage, deterioration, and wear — Oil leakage • Are all the items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
9	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH MASTER CYLINDER** • Inspect the following of the clutch master cylinder. (See CLUTCH MASTER CYLINDER REMOVAL/ INSTALLATION [F65M-R].) — Installation condition of part — Damage, deterioration, and wear — Cylinder movement • Are all the items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
10	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH RELEASE CYLINDER** • Inspect the following of the clutch release cylinder. (See CLUTCH RELEASE CYLINDER REMOVAL/ INSTALLATION [F65M-R].) — Installation condition of part — Damage, deterioration, and wear — Cylinder movement • Are all the items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
11	**INSPECT IF MALFUNCTION CAUSE IS RELEASE FORK** • Inspect the following of the release fork. (See CLUTCH UNIT REMOVAL/INSTALLATION [F65M-R].) — Installation condition of part — Damage, deterioration, and wear — Pivot pin damage, deterioration, and wear • Is the release fork normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.

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12	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH RELEASE COLLAR** • Inspect the clutch release collar. (See CLUTCH RELEASE COLLAR INSPECTION [F65M-R].) • Is the clutch release collar normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
13	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH COVER** • Inspect the clutch cover. (See CLUTCH COVER INSPECTION [F65M-R].) • Is the clutch cover normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
14	**INSPECT IF MALFUNCTION CAUSE IS CLUTCH DISC** • Inspect the clutch disc. (See CLUTCH DISC INSPECTION [F65M-R].) • Is the clutch disc normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
15	**INSPECT IF MALFUNCTION CAUSE IS FLYWHEEL** • Inspect the flywheel. (See FLYWHEEL INSPECTION [F65M-R].) • Is the flywheel normal?	Yes	Manual transaxle internal malfunction can be considered. • Replace the manual transaxle. (See MANUAL TRANSAXLE REMOVAL/ INSTALLATION [F65M-R].)
		No	Repair or replace the malfunctioning location and perform the repair completion verification.
Repair completion verification	**VERIFY THAT VEHICLE IS REPAIRED** • Install/connect the part removed/disconnected during the troubleshooting procedure. • Has the malfunction symptom been eliminated?	Yes	Complete the symptom troubleshooting. (Explain contents of repair to customer)
		No	Repeat the diagnosis from Step 1.